THE JACCARD TRIANGLE INEQUALITY FOR SUPERMODULAR LOG-SUBMODULAR VALUATIONS ON ARBITRARY LATTICES

ABSTRACT. Let L be a lattice and $f\colon L \to (0,\infty)$ a valuation. The generalized Jaccard distance is $d_{f,J}(A,B) = 1 - f(A \wedge B)/f(A \vee B)$. Badica and Badica prove that $d_{f,J}$ satisfies the triangle inequality when f is strictly positive, monotone, supermodular and log-submodular, provided L is relatively complemented and distributive, and ask in their Section 7 whether those two structural hypotheses on L can be removed. We observe that they can: the triangle inequality holds on an arbitrary lattice, with no distributivity, relative complementation, bounds or finiteness assumed. The proof applies supermodularity once and log-submodularity once, not to A, B, C but to the derived pairs $(A \wedge B,\ B \wedge C)$ and $(A \vee B,\ B \vee C)$, and collapses to the inequality $(P-p)(Q-q) \geq (b-p)(b-q)$ between five values of f. Dropping either supermodularity or log-submodularity breaks the conclusion already on the four-element Boolean lattice.

1. THE QUESTION

Let L be a lattice, written with $\wedge$ and $\vee$ and ordered by $X \leq Y \iff X \wedge Y = X$. Call $f\colon L \to (0,\infty)$ *monotone* if $X \leq Y \Rightarrow f(X) \leq f(Y)$, *supermodular* if

$$f(X) + f(Y) \ \leq\ f(X \vee Y) + f(X \wedge Y) \qquad (X, Y \in L), \tag{1}$$

and *log-submodular* if

$$f(X)\, f(Y) \ \geq\ f(X \vee Y)\, f(X \wedge Y) \qquad (X, Y \in L). \tag{2}$$

The *generalized Jaccard distance* of f is

$$d_{f,J}(A,B) \ =\ 1 - \frac{f(A \wedge B)}{f(A \vee B)}, \qquad S(A,B) \ :=\ \frac{f(A \wedge B)}{f(A \vee B)}, \tag{3}$$

so $d_{f,J} = 1 - S$, and the triangle inequality $d_{f,J}(A,C) \leq d_{f,J}(A,B) + d_{f,J}(B,C)$ is exactly

$$S(A,B) + S(B,C) \ \leq\ 1 + S(A,C). \tag{4}$$

In the preprint [1], Badica and Badica prove (4) for strictly positive monotone supermodular log-submodular f on a lattice that is relatively complemented *and* distributive (their supermodular log-submodular theorem), and

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separately prove (4) for modular f on an arbitrary lattice (their modular theorem). Their Section 7, *Discussion of Open Theoretical Problems*, lists four open problems; the second of them reads verbatim and in full:

> **Arbitrary lattices for supermodular valuations:** While we established metricity for supermodular and log-submodular valuations, the proof relies on the lattice being both relatively complemented and distributive. It is still unknown whether the standard generalized Jaccard formula holds up to the triangle inequality for these valuations on arbitrary lattices that lack those two specific traits.

Here "the standard generalized Jaccard formula" is (3) and "these valuations" is the hypothesis class of their supermodular log-submodular theorem, namely strictly positive, monotone, supermodular and log-submodular. No conjecture is attached to the bullet, so what follows answers it in the affirmative and neither confirms nor refutes a stated guess.

2. The theorem

Theorem 1. *Let L be a lattice, not assumed distributive, relatively complemented, bounded or finite, and let $f\colon L \to (0,\infty)$ be monotone, supermodular and log-submodular. Then $d_{f,J}$ satisfies the triangle inequality: for all $A, B, C \in L$,*

$$d_{f,J}(A,C) \ \le\ d_{f,J}(A,B) + d_{f,J}(B,C).$$

Hence the supermodular log-submodular theorem of [1] holds verbatim with the words "relatively complemented distributive" deleted, which answers the bullet quoted above.

Proof. Fix $A, B, C \in L$ and abbreviate the seven values of f involved:

$$p = f(A \wedge B), \quad P = f(A \vee B), \quad q = f(B \wedge C), \quad Q = f(B \vee C),$$

$$r = f(A \wedge C), \quad R = f(A \vee C), \quad b = f(B).$$

All seven are > 0. Since $A \wedge B \le B \le A \vee B$ and $B \wedge C \le B \le B \vee C$, monotonicity gives

$$0 < p \le b \le P, \qquad 0 < q \le b \le Q. \tag{5}$$

By (4) it suffices to prove $p/P + q/Q \le 1 + r/R$.

Step 1 (supermodularity, on the pair of meets). Put $X = A \wedge B$ and $Y = B \wedge C$. Both lie below B, so $X \vee Y \le B$; and $X \wedge Y = A \wedge B \wedge C \le A \wedge C$. Applying (1) to this pair and then monotonicity twice,

$$p + q \ =\ f(X) + f(Y) \ \le\ f(X \vee Y) + f(X \wedge Y) \ \le\ b + r,$$

that is

$$r \ \ge\ p + q - b. \tag{6}$$

Step 2 (log-submodularity, on the pair of joins). Put $X = A \vee B$ and $Y = B \vee C$. Both lie above B, so $X \wedge Y \geq B$; and $X \vee Y = A \vee B \vee C \geq A \vee C$. Applying (2) to this pair and then monotonicity twice,

$$PQ \;=\; f(X)f(Y) \;\geq\; f(X \vee Y)\, f(X \wedge Y) \;\geq\; R\, b,$$

that is

$$R \;\leq\; \frac{PQ}{b}. \tag{7}$$

Step 3 (a lower bound for $S(A, C)$). We claim $r/R \;\geq\; b(p+q-b)/(PQ)$. If $p + q - b \leq 0$ the right-hand side is $\leq 0 < r/R$. If $p + q - b > 0$ then $r \geq p+q-b > 0$ by (6) and $0 < R \leq PQ/b$ by (7), so $r/R \geq (p+q-b)\, b/(PQ)$. In either case it suffices to prove

$$\frac{p}{P} + \frac{q}{Q} \;\leq\; 1 + \frac{b\,(p + q - b)}{PQ}. \tag{8}$$

Step 4 (clearing denominators). Multiplying (8) by $PQ > 0$ turns it into $pQ + qP \leq PQ + bp + bq - b^2$, and

$$\big(pQ + qP\big) - \big(PQ + bp + bq - b^2\big) \;=\; -\Big[(P - p)(Q - q) - (b - p)(b - q)\Big]$$

identically in p, P, q, Q, b — both sides expand to $pQ+Pq-PQ-bp-bq+b^2$. So (8) is precisely

$$(P - p)(Q - q) \;\geq\; (b - p)(b - q). \tag{9}$$

Step 5 (a termwise certificate for (9)). Identically in p, P, q, Q, b,

$$(P - p)(Q - q) - (b - p)(b - q) \;=\; (P - b)(Q - q) + (b - p)(Q - b). \tag{10}$$

By (5) each of $P - b$, $Q - q$, $b - p$, $Q - b$ is ≥ 0, so the right-hand side of (10) is a sum of two products of non-negative numbers and is ≥ 0. This proves (9), hence (8), hence (4). □

Remark 2. The proof uses only the four inequalities (5) and one application each of (1) and (2), to the derived pairs $(A \wedge B,\ B \wedge C)$ and $(A \vee B,\ B \vee C)$ rather than to (A, B), (B, C) or (A, C). The proof in [1] instead constructs relative complements inside the interval $[A \wedge B \wedge C,\ A \vee B \vee C]$ and then applies distributivity, which is where its two structural hypotheses on L enter.

Remark 3. Theorem 1 is about the triangle inequality, which is what the bullet asks about. It does not upgrade $d_{f,J}$ to a metric: $d_{f,J}(X, Y) = 0$ exactly when f is constant on $[X \wedge Y,\ X \vee Y]$, so $d_{f,J}$ is a pseudometric in general and a metric exactly when f is strictly monotone. The same caveat attaches to the theorems of [1].

3. Both hypotheses on f are needed

Write B_2 for the four-element Boolean lattice, given by its covers:

$$B_2 : \quad 0 < x, \quad 0 < y, \quad x < 1, \quad y < 1. \tag{11}$$

Write a valuation on B_2 as the tuple $(f(0), f(x), f(y), f(1))$, and recall $\Delta(A, B, C) := S(A, B) + S(B, C) - 1 - S(A, C)$, so that $\Delta > 0$ is a violation of (4). Each row below is an exact rational computation on the object printed in (11).

f	lost	(A, B, C)	Δ	step that breaks
$(2, 3, 3, 20)$	(2)	$(x, 0, y)$	$\frac{7}{30}$	2: $Rb = 40 > 9 = PQ$
$(1, 3, 3, 4)$	(1)	$(x, 1, y)$	$\frac{1}{4}$	1: $p + q - b = 2 > 1 = r$

The first valuation is monotone and supermodular but not log-submodular ($f(1)f(0) = 40 > 9 = f(x)f(y)$); the second is monotone and log-submodular but not supermodular ($f(1) + f(0) = 5 < 6 = f(x) + f(y)$). So neither supermodularity nor log-submodularity may be dropped from Theorem 1, and neither is implied by monotonicity; and since B_2 is distributive and relatively complemented, this bounds the hypotheses on f and not the hypotheses on L that Theorem 1 removes.

Theorem 1 is a sufficiency statement. Section 6 of [1] proves, on an arbitrary lattice, that monotonicity together with supermodularity of f *and* of $1/f$ is *necessary* for (4); that is a different hypothesis class, so Theorem 1 is neither the converse of it nor half of a characterisation. The remaining open problems of Section 7 of [1] are untouched here.

4. Verification

The proof of Theorem 1 is a hand argument and needs no machine. The accompanying program `verify.py` (Python 3.9 or later, standard library only; exact arithmetic throughout, no floating point) re-derives the numbers stated here: it verifies the two polynomial identities of Steps 4 and 5 as exact identities in $\mathbb{Z}[p, P, q, Q, b]$ by sparse expansion rather than by sampling; it confirms the four order facts used in Steps 1 and 2 at every ordered triple of every lattice of order at most six; it checks that the covers (11) define a lattice; and it reproduces the two violation margins 7/30 and 1/4 tabulated above together with the step each breaks. It prints one line per check and closes with `VERDICT: ALL 27 CHECKS PASS`, exiting 0 only if all pass.

Most of the 27 checks bear no load on Theorem 1: finite sweeps over small lattices, an enumerator control and further example valuations are recorded but are not used above, and no program can quantify over the class the theorem covers. The recorded output is also not labelled for the present text: it letters its own sections A–G, calls the inequalities (6) $r \geq p + q - b$ and (7) $R \leq PQ/b$ of Steps 1 and 2 *Lemma 1* and *Lemma 2*, and describes three lattices, three totals and a valuation g as printed in Sections 3 and 4 of the paper. None of those cross-references holds here: the only objects this paper prints are the lattice (11) and the two valuations tabulated in Section 3, it states no totals, and the valuation g appears only in the program. The theorems of [1] are quoted there, not re-proved.

References

[1] C. Badica and A. Badica, *On the triangle inequality for the Jaccard distance in arbitrary lattices*, arXiv:2608.18194v1, 2026. arXiv:2608.18194. The passages quoted above are from the v1 e-print, the only version available at the time of writing.